BIRZEIT UNIVERSITY

*Faculty of Engineering and Technology*

*Electrical and Computer Engineering Department*

# *Artificial Intelligence*

## *ENCS3340*

Project 2


Prepared by:

**Mohammad Omar** **1221332**

**Heba Mustafa** **1221916**


Instructor: Dr. Yazan Abu Farha

Section: 1

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## Table of Content

Table of Content ..... I
1. Introduction ..... 1
2. Dataset Description ..... 2
3. Model Descriptions ..... 3
3.1 Naive Bayes Classifier ..... 3
3.2 Decision Tree Classifier ..... 3
3.3 Feedforward Neural Network (MLP Classifier) ..... 3
4. Evaluation Criteria ..... 3
5. Results ..... 4
6. Comparative Analysis ..... 5
(Extra) Screenshots for Full Run ..... 6

## 1. Introduction

This project compares three machine learning methods: Naive Bayes, Decision Tree, and a Feedforward Neural Network (MLP) on image classification. A balanced pool of more than 500 pictures, evenly spread across three classes, was resized to 64 by 64 pixels and turned into one-dimensional vectors before training. The aim is to judge each technique by how well it labels images, how easy its decisions are to explain, and the time and memory it consumes. Naive Bayes, praised for its speed and straightforward math, sets the first benchmark. The tree, with its human-readable branches, offers clear rules that analysts can follow. The MLP digs deeper into the pixel data, learning complex, non-linear patterns that the other two might miss. Performance is measured through accuracy, precision, recall, F1-score, and confusion matrices so readers can see failures as well as successes. Results confirm the familiar trade-off; the MLP usually wins on raw accuracy, yet the lighter models still matter because they run faster and tell clear stories. These findings help teams choose the right tool for classifying images when speed, insight, or sheer prediction power is the most urgent concern.

## 2. Dataset Description

The dataset consists of color images taken from various online sources such as Kaggle, arranged into three distinct fruit categories: **blueberries, pomegranates, and bananas**. To ensure balance and fairness in model comparisons, we selected an equal number of images from each category: 249 images per category (747 images in total).

1. Image size: Each image was resized to 64 x 64 pixels.
2. Color channels: RGB (3 channels)
3. Preprocessing: Images were resized and converted to RGB.
4. The resulting feature vectors were flattened for input into traditional classifiers

Figure 1: Dataset Categories

## 3. Model Descriptions

### 3.1 Naive Bayes Classifier

Naive Bayes operates on probabilities, based on Bayes' rule. It treats each feature as if it were working alone, so it works quickly on large inputs such as image networks. For this task, we chose scikit-learns Gaussian NB because our data is often connected.

- Pros: Very fast, requires little memory.
- Cons: The independence story rarely matches the actual pixels.

### 3.2 Decision Tree Classifier

A decision tree travels a yes/no path through the data, starting from the root. It takes features directly from the CNN and partitions them feature by feature. We return to scikit-learns Decision Tree Classifier.

- Pros: Easy to read, can be plotted.
- Cons: Preserves noise if the tree gets too deep.

### 3.3 Feedforward Neural Network (MLP Classifier)

MLP, or Multi-Layer Perceptron, is a simple feedforward network built using the MLP Classifier sklearns. It hides one or more layers between the input and output. We tested one hidden layer with 128 neurons.

All inputs are CNN vectors scaled using Standard Scaler.

- Advantages: Learns sharp and curved decision boundaries.
- Cons: Requires precise control buttons and takes longer to train.

## 4. Evaluation Criteria

We evaluated every model with these straightforward measures:

1. Accuracy: Share of correct guesses out of all made.
2. Precision: Hits divided by hits plus the false alarms.
3. Recall: Hits stated divided by hits missed.
4. F1-score: The tidy average of precision and recall.
5. Confusion matrix: A table that counts each class correct and wrong calls.

Note: All tests ran on a 20 percent hold-out split of the balanced data.

## 5. Results

### Final Results Summary

| 88.93% | 77.07% | 98.00% |
|---|---|---|
| DECISION TREE | NAIVE BAYES | CNN + MLP |

### Decision Tree

| Class | Precision | Recall | F1-Score | Support |
|---|---|---|---|---|
| Banana | 0.898 | 0.880 | 0.889 | 50 |
| Blueberries | 0.904 | 0.940 | 0.922 | 50 |
| Pomegranates | 0.878 | 0.860 | 0.869 | 50 |
| Macro Avg | 0.893 | 0.893 | 0.893 | 150 |
| Weighted Avg | 0.893 | 0.893 | 0.893 | 150 |

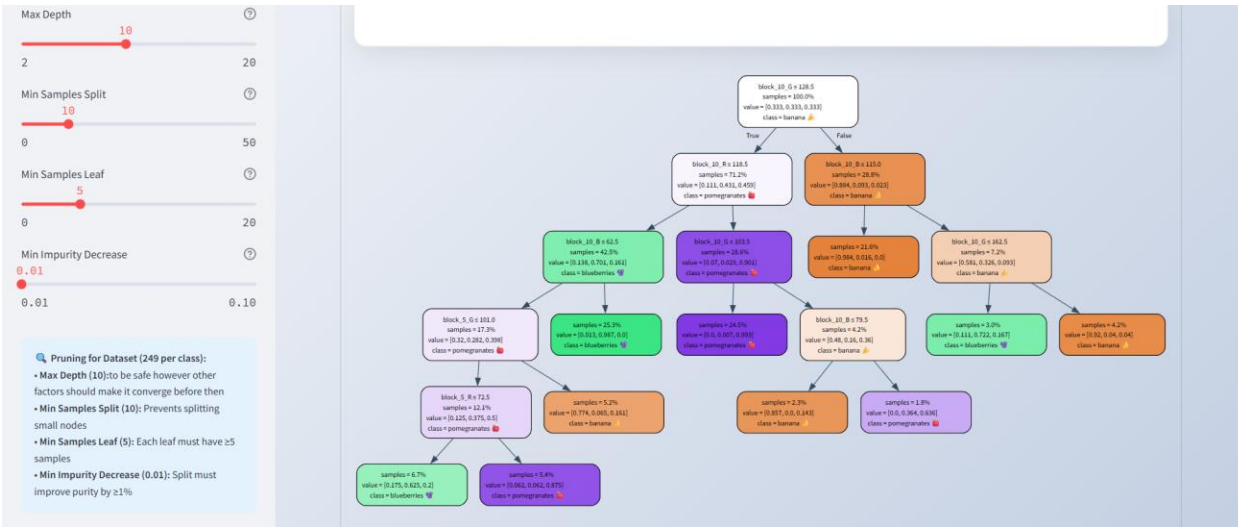
### Naive Bayes

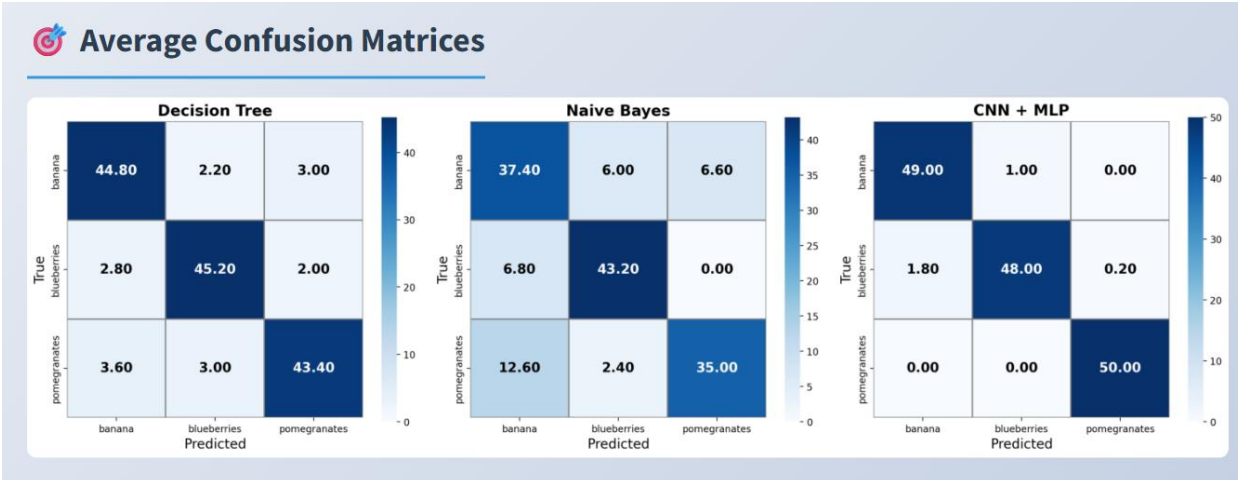
| Class | Precision | Recall | F1-Score | Support |
|---|---|---|---|---|
| Banana | 0.644 | 0.760 | 0.697 | 50 |
| Blueberries | 0.816 | 0.800 | 0.808 | 50 |
| Pomegranates | 0.857 | 0.720 | 0.783 | 50 |
| Macro Avg | 0.773 | 0.760 | 0.763 | 150 |
| Weighted Avg | 0.773 | 0.760 | 0.763 | 150 |

### CNN + MLP

| Class | Precision | Recall | F1-Score | Support |
|---|---|---|---|---|
| Banana | 0.942 | 0.980 | 0.961 | 50 |
| Blueberries | 0.979 | 0.940 | 0.959 | 50 |
| Pomegranates | 1.000 | 1.000 | 1.000 | 50 |
| Macro Avg | 0.974 | 0.973 | 0.973 | 150 |
| Weighted Avg | 0.974 | 0.973 | 0.973 | 150 |

Please, Run the code




## 6. Comparative Analysis

The naive Bayes algorithm was fast and simple, but its performance was poor due to its assumption of feature independence.

The decision tree improved this performance, but showed signs of overfitting, especially when depth was not constrained.

MLP Classifier clearly outperformed both, leveraging its nonlinear transformation capabilities and rich features in convolutional neural networks (CNNs). However, its training time was longer.

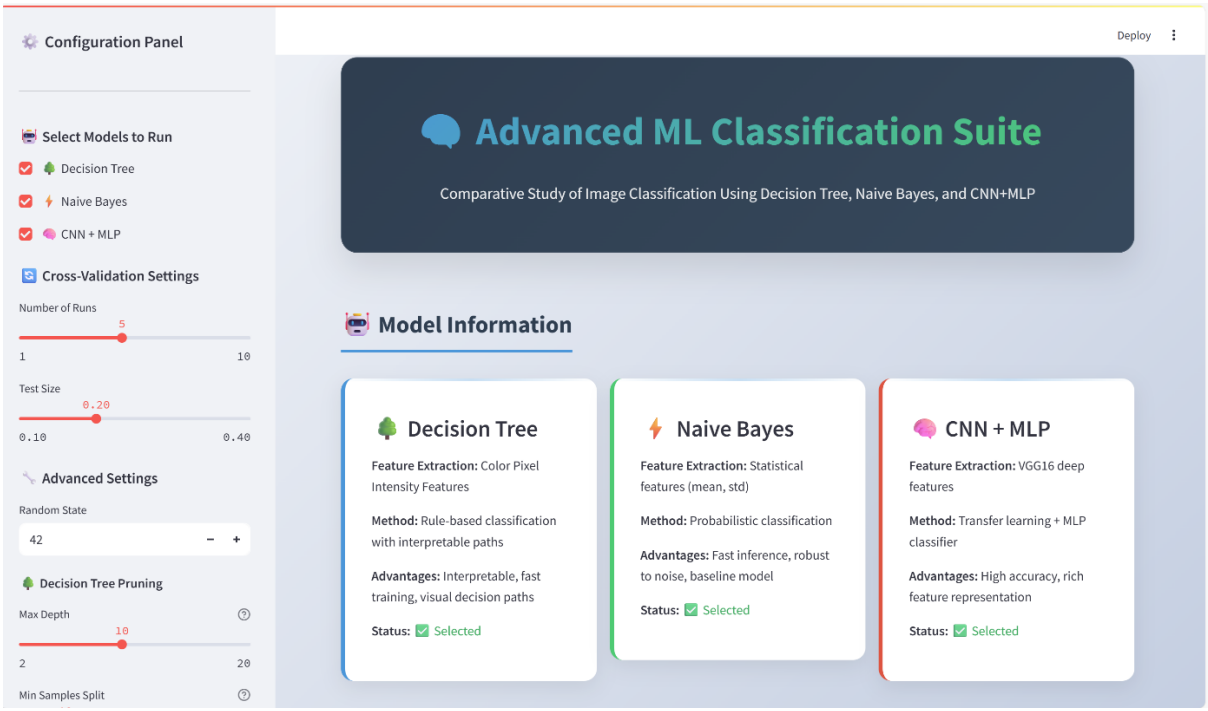
We can summarize the Comparative analysis in the following table: -

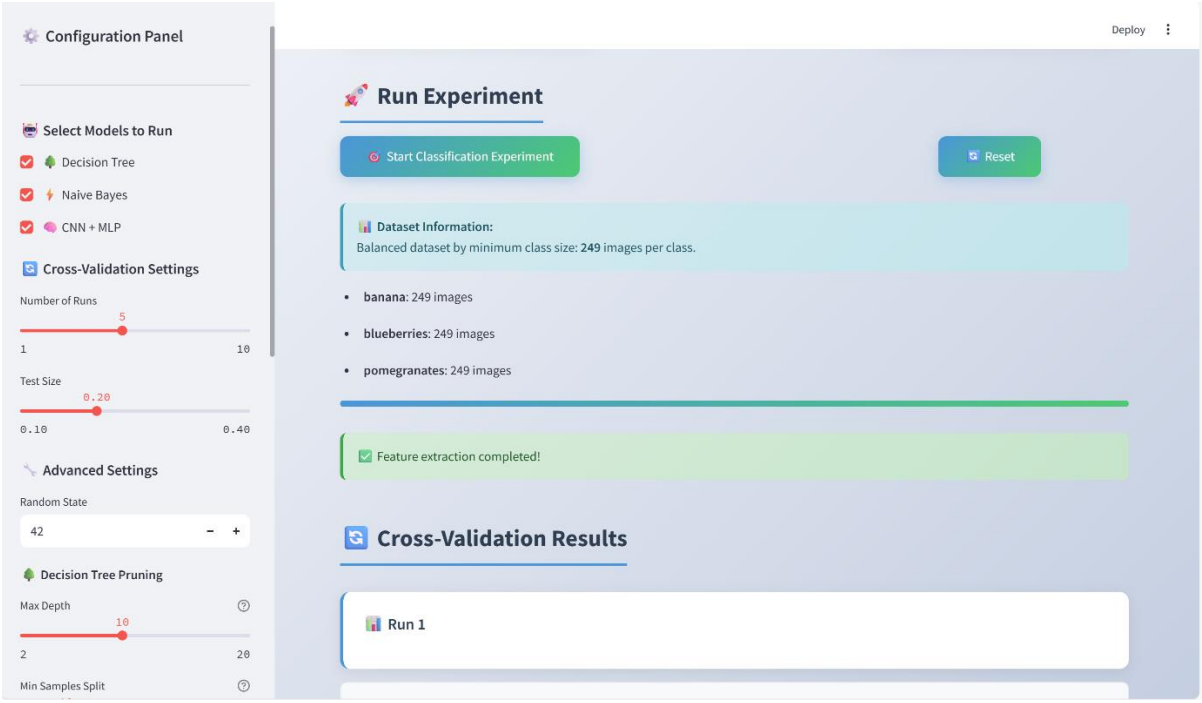
Table 1: Comparative analysis

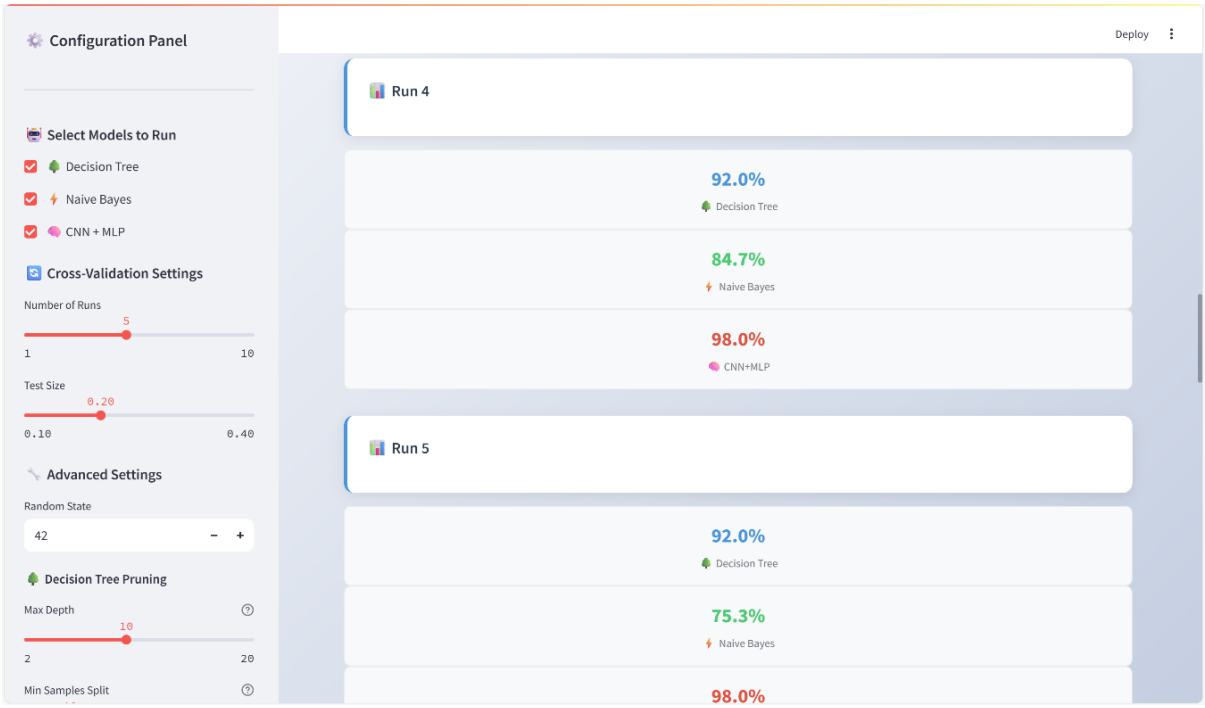
| Model | Training Time | Strengths | Weaknesses |
|---|---|---|---|
| **Naive Bayes** | Very Fast | Simple, fast, low memory usage | Poor performance due to feature independence |
| **Decision Tree** | Fast | Interpretable, captures patterns | Prone to overfitting |
| **MLP (1 hidden layer)** | long | Captures nonlinear features well | Longer training time |

Note: we run the code many times to take the average accuracy and then we build our comparative analysis based on it.

## (Extra) Screenshots for Full Run




Configuration Panel
Deploy
Select Models to Run
Decision Tree
Naive Bayes
CNN + MLP
Cross-Validation Settings
Number of Runs
5
1
10
Test Size
0.20
0.10
0.40
Advanced Settings
Random State
42
Decision Tree Pruning
Max Depth
10
2
20
Min Samples Split
Run Experiment
Start Classification Experiment
Reset
Dataset Information:
Balanced dataset by minimum class size: 249 images per class.
banana: 249 images
blueberries: 249 images
pomegranates: 249 images
Feature extraction completed!
Cross-Validation Results
Run 1


Configuration Panel
Deploy
Select Models to Run
Decision Tree
Naive Bayes
CNN + MLP
Cross-Validation Settings
Number of Runs
5
1
10
Test Size
0.20
0.10
0.40
Advanced Settings
Random State
42
Decision Tree Pruning
Max Depth
10
2
20
Min Samples Split
Run 4
92.0%
Decision Tree
84.7%
Naive Bayes
98.0%
CNN+MLP
Run 5
92.0%
Decision Tree
75.3%
Naive Bayes
98.0%

Configuration Panel

Select Models to Run

- Decision Tree
- Naive Bayes
- CNN + MLP

Cross-Validation Settings

Number of Runs: 5 (1 – 10)

Test Size: 0.20 (0.10 – 0.40)

Advanced Settings

Random State: 42

Decision Tree Pruning

Max Depth: 10 (2 – 20)

Min Samples Split

Deploy

## Final Results Summary

| Decision Tree | Naive Bayes | CNN + MLP |
|---|---|---|
| 89.20% | 77.87% | 98.53% |

## Classification Reports

### Decision Tree

| Class | Precision | Recall | F1-Score | Support |
|---|---|---|---|---|
| Banana | 0.920 | 0.920 | 0.920 | 50 |
| Blueberries | 0.906 | 0.960 | 0.932 | 50 |
| Pomegranates | 0.936 | 0.880 | 0.907 | 50 |
| Macro Avg | 0.921 | 0.920 | 0.920 | 150 |
| Weighted Avg | 0.921 | 0.920 | 0.920 | 150 |

### Naive Bayes

| Class | Precision | Recall | F1-Score | Support |
|---|---|---|---|---|

Configuration Panel

Select Models to Run

- Decision Tree
- Naive Bayes
- CNN + MLP

Cross-Validation Settings

Number of Runs: 5 (1 – 10)

Test Size: 0.20 (0.10 – 0.40)

Advanced Settings

Random State: 42

Decision Tree Pruning

Max Depth: 10

Deploy

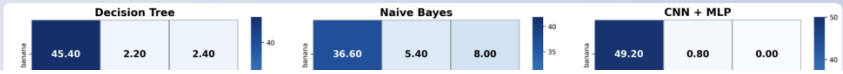
### Naive Bayes

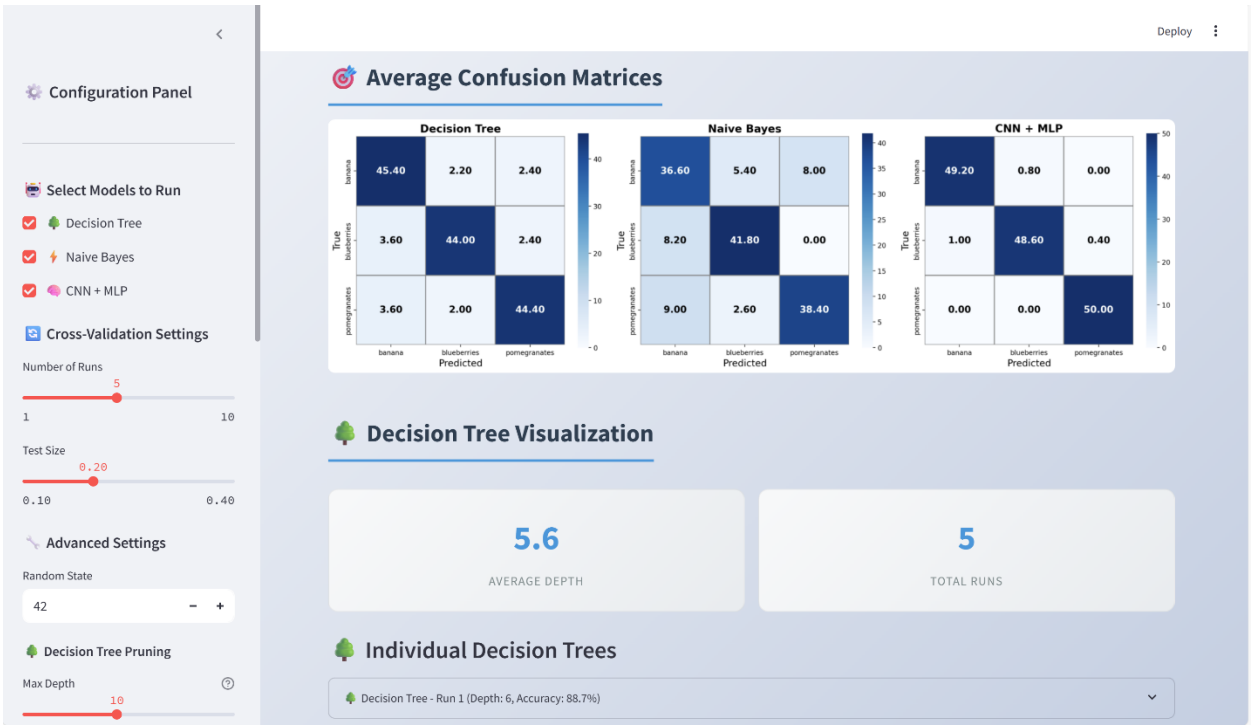
| Class | Precision | Recall | F1-Score | Support |
|---|---|---|---|---|
| Banana | 0.625 | 0.700 | 0.660 | 50 |
| Blueberries | 0.889 | 0.800 | 0.842 | 50 |
| Pomegranates | 0.776 | 0.760 | 0.768 | 50 |
| Macro Avg | 0.763 | 0.753 | 0.757 | 150 |
| Weighted Avg | 0.763 | 0.753 | 0.757 | 150 |

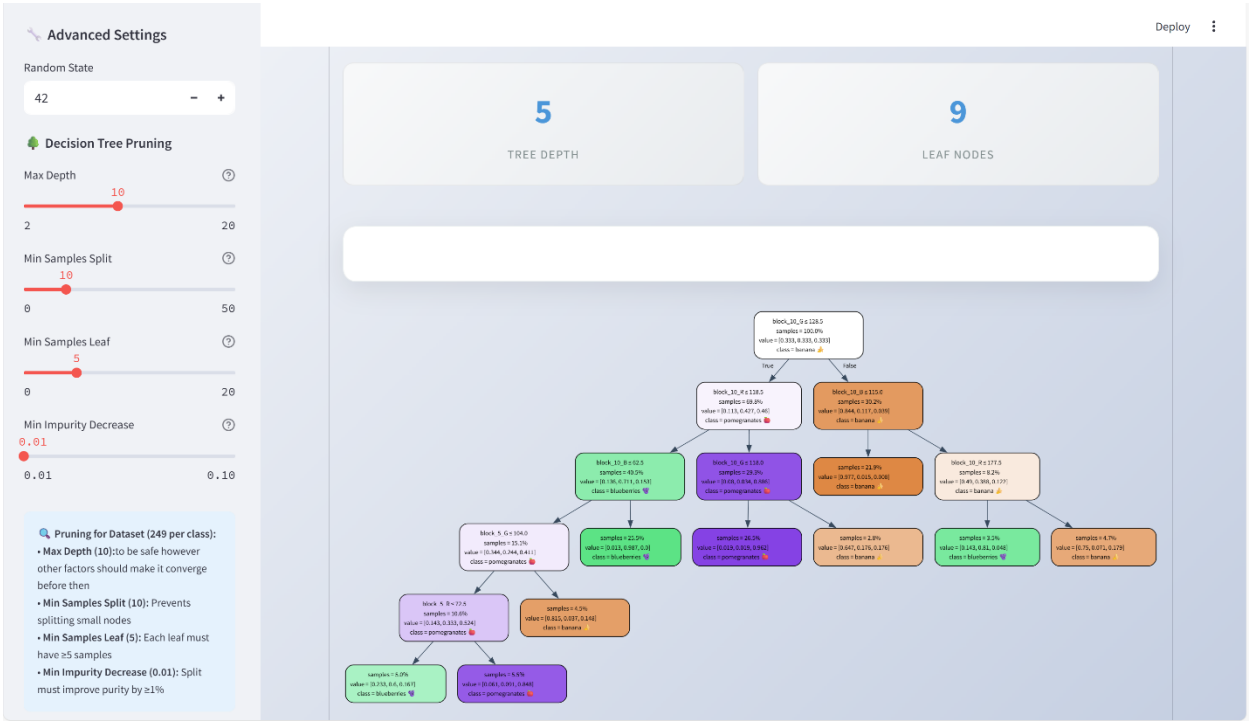
### CNN + MLP

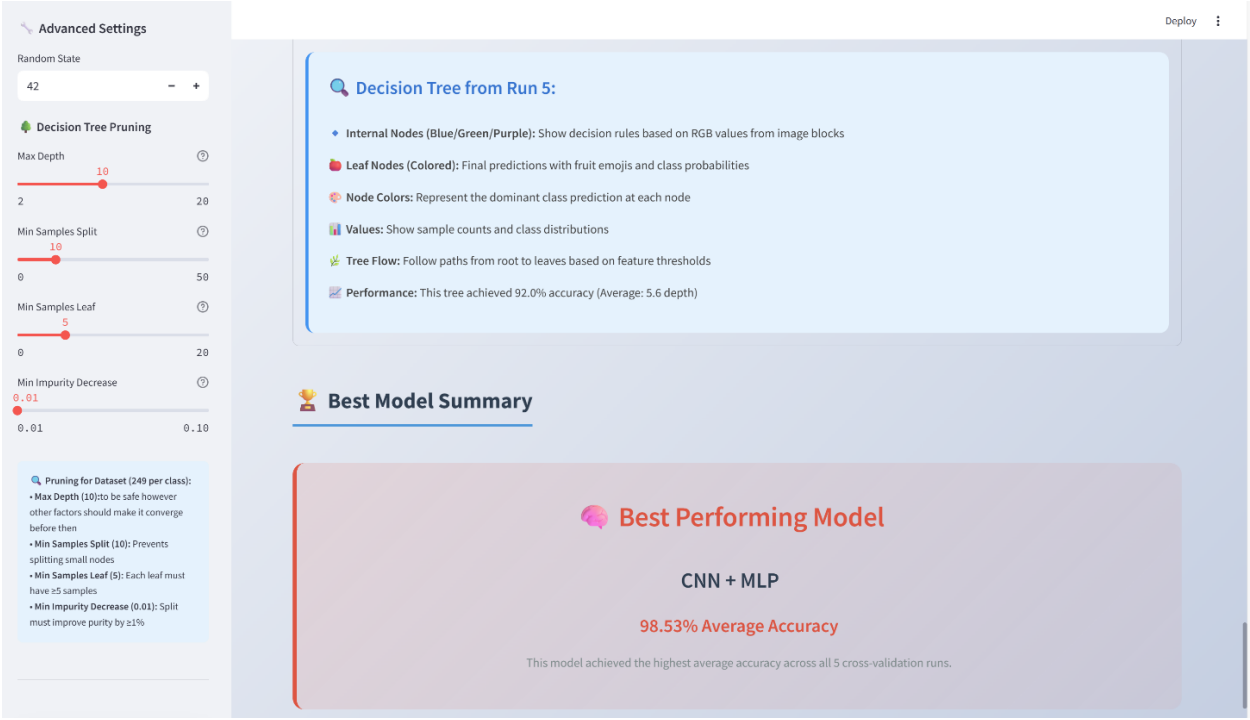
| Class | Precision | Recall | F1-Score | Support |
|---|---|---|---|---|
| Banana | 0.961 | 0.980 | 0.970 | 50 |
| Blueberries | 0.980 | 0.960 | 0.970 | 50 |
| Pomegranates | 1.000 | 1.000 | 1.000 | 50 |
| Macro Avg | 0.980 | 0.980 | 0.980 | 150 |
| Weighted Avg | 0.980 | 0.980 | 0.980 | 150 |

## Average Confusion Matrices




Deploy
Configuration Panel
Select Models to Run
Decision Tree
Naive Bayes
CNN + MLP
Cross-Validation Settings
Number of Runs
5
1
10
Test Size
0.20
0.10
0.40
Advanced Settings
Random State
42
Decision Tree Pruning
Max Depth
10
Average Confusion Matrices
Decision Tree
45.40 2.20 2.40
3.60 44.00 2.40
3.60 2.00 44.40
Naive Bayes
36.60 5.40 8.00
8.20 41.80 0.00
9.00 2.60 38.40
CNN + MLP
49.20 0.80 0.00
1.00 48.60 0.40
0.00 0.00 50.00
banana
blueberries
pomegranates
True
Predicted
Decision Tree Visualization
5.6
AVERAGE DEPTH
5
TOTAL RUNS
Individual Decision Trees
Decision Tree - Run 1 (Depth: 6, Accuracy: 88.7%)


Deploy
Advanced Settings
Random State
42
Decision Tree Pruning
Max Depth
10
2
20
Min Samples Split
10
0
50
Min Samples Leaf
5
0
20
Min Impurity Decrease
0.01
0.01
0.10
Pruning for Dataset (249 per class):
• Max Depth (10):to be safe however other factors should make it converge before then
• Min Samples Split (10): Prevents splitting small nodes
• Min Samples Leaf (5): Each leaf must have ≥5 samples
• Min Impurity Decrease (0.01): Split must improve purity by ≥1%
5
TREE DEPTH
9
LEAF NODES




Note:

We have tree pruning which is Preprocessing Pruning (Min Samples Split, Min Samples Leaf, and Min Impurity Decrease) they limit the splitting during training, which prevents overfitting and a complex tree.